

```

from google.colab import drive
drive.mount('/content/drive/')

Mounted at /content/drive/

!pip install SentencePiece
!pip install transformers
!pip install googletrans==3.1.0a0

Downloading safetensors-0.3.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.3 MB)
1.3/1.3 MB 19.0 MB/s eta 0:00:00
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.10/dist-packages (from transformers) (4.66.1)
Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages (from huggingface-hub<1.0,>=0.15.1->trai
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.10/dist-packages (from huggingface-hu
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests->transi
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests->transformers) (3.4
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests->transformers
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests->transformers
Installing collected packages: tokenizers, safetensors, huggingface-hub, transformers
Successfully installed huggingface-hub-0.16.4 safetensors-0.3.3 tokenizers-0.13.3 transformers-4.33.1
Collecting googletrans==3.1.0a0
Downloading googletrans-3.1.0a0.tar.gz (19 kB)
Preparing metadata (setup.py) ... done
Collecting httpx==0.13.3 (from googletrans==3.1.0a0)
Downloading httpx-0.13.3-py3-none-any.whl (55 kB)
55.1/55.1 kB 2.0 MB/s eta 0:00:00
Requirement already satisfied: certifi in /usr/local/lib/python3.10/dist-packages (from httpx==0.13.3->googletrans==3.1.0
Collecting hstspreload (from httpx==0.13.3->googletrans==3.1.0a0)
Downloading hstspreload-2023.1.1-py3-none-any.whl (1.5 MB)
1.5/1.5 MB 3.7 MB/s eta 0:00:00
Requirement already satisfied: sniffio in /usr/local/lib/python3.10/dist-packages (from httpx==0.13.3->googletrans==3.1.0
Collecting chardet==3.* (from httpx==0.13.3->googletrans==3.1.0a0)
Downloading chardet-3.0.4-py2.py3-none-any.whl (133 kB)
133.4/133.4 kB 9.5 MB/s eta 0:00:00
Collecting idna==2.* (from httpx==0.13.3->googletrans==3.1.0a0)
Downloading idna-2.10-py2.py3-none-any.whl (58 kB)
58.8/58.8 kB 6.1 MB/s eta 0:00:00
Collecting rfc3986<2,>=1.3 (from httpx==0.13.3->googletrans==3.1.0a0)
Downloading rfc3986-1.5.0-py2.py3-none-any.whl (31 kB)
Collecting httpcore==0.9.* (from httpx==0.13.3->googletrans==3.1.0a0)
Downloading httpcore-0.9.1-py3-none-any.whl (42 kB)
42.6/42.6 kB 2.7 MB/s eta 0:00:00
Collecting h11<0.10,>=0.8 (from httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
Downloading h11-0.9.0-py2.py3-none-any.whl (53 kB)
53.6/53.6 kB 3.4 MB/s eta 0:00:00
Collecting h2==3.* (from httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
Downloading h2-3.2.0-py2.py3-none-any.whl (65 kB)
65.0/65.0 kB 5.4 MB/s eta 0:00:00
Collecting hyperframe<6,>=5.2.0 (from h2==3.*->httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
Downloading hyperframe-5.2.0-py2.py3-none-any.whl (12 kB)
Collecting hpack<4,>=3.0 (from h2==3.*->httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
Downloading hpack-3.0.0-py2.py3-none-any.whl (38 kB)
Building wheels for collected packages: googletrans
Building wheel for googletrans (setup.py) ... done
Created wheel for googletrans: filename=googletrans-3.1.0a0-py3-none-any.whl size=16352 sha256=4038a2ceff3927f40183d976
Stored in directory: /root/.cache/pip/wheels/50/5d/3c/8477d0af4ca2b8b1308812c09f1930863caeebc762fe265a95
Successfully built googletrans
Installing collected packages: rfc3986, hyperframe, hpack, h11, chardet, idna, hstspreload, h2, httpcore, httpx, googletr
Attempting uninstall: chardet
Found existing installation: chardet 5.2.0
Uninstalling chardet-5.2.0:
Successfully uninstalled chardet-5.2.0
Attempting uninstall: idna
Found existing installation: idna 3.4
Uninstalling idna-3.4:
Successfully uninstalled idna-3.4
Successfully installed chardet-3.0.4 googletrans-3.1.0a0 h11-0.9.0 h2-3.2.0 hpack-3.0.0 hstspreload-2023.1.1 httpcore-0.9.1 httpx-0.13.3

import googletrans
googletrans.__version__

'3.1.0-alpha'

import os
import pandas as pd
from transformers import PegasusTokenizer, PegasusForConditionalGeneration, TFPegasusForConditionalGeneration, pipeline
import googletrans
from googletrans import Translator
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import torch

import pandas as pd

```

```
df = pd.read_excel("/content/drive/MyDrive/Mortoga project/test_texts.xlsx")
df = df[0:10]
df
```

	ID	TEXT	source
0	1	In einem Krankenhaus in Canberra haben Ärzte e...	https://www.sueddeutsche.de/wissen/australien-...
1	2	Künstliche Intelligenz (KI) findet auch in der...	https://www.bild.de/ratgeber/gesundheit/gesund...
2	3	Das landläufige Bild, das der Schriftsteller W...	https://www.jungewelt.de/artikel/457367.litera...
3	4	In Kanada organisieren sich Drogenkonsument*in...	https://www.nd-aktuell.de/artikel/1175963.drog...
4	5	taz: Frau Scharrel, immer wieder werden Frauen...	https://taz.de/Verbandschefin-ueber-Abtreibung...
5	6	Wer Hundebesitzer im Umgang mit ihren Tieren b...	https://www.tagesspiegel.de/gesundheit/von-weg...

▼ Translating to English

```
def translate(text):
    translator = Translator()
    return translator.translate(text, src='de', dest='en').text
```

```
df['English Translated Text'] = df['TEXT'].apply(translate)
df
```

	ID	TEXT	source	English Translated Text
0	1	In einem Krankenhaus in Canberra haben Ärzte e...	https://www.sueddeutsche.de/wissen/australien-...	Doctors removed a live roundworm, three inches...
1	2	Künstliche Intelligenz (KI) findet auch in der...	https://www.bild.de/ratgeber/gesundheit/gesund...	Artificial intelligence (AI) is also finding m...
2	3	Das landläufige Bild, das der Schriftsteller W...	https://www.jungewelt.de/artikel/457367.litera...	The common image that the writer Wolfgang Herr...
3	4	In Kanada organisieren sich Drogenkonsument*in...	https://www.nd-aktuell.de/artikel/1175963.drog...	In Canada, drug users organize themselves in a...
4	5	taz: Frau Scharrel, immer wieder werden Frauen...	https://taz.de/Verbandschefin-ueber-Abtreibung...	taz: Ms. Scharrel, gynecologists who offer abo...
5	6	Wer Hundebesitzer im Umgang mit ihren Tieren b...	https://www.tagesspiegel.de/gesundheit/von-weg...	Anyone who observes dog owners interacting wit...
		Impfstoffe auf Basis von mRNA brachten im		Vaccines based on mRNA brought the

▼ Using Finetuned Pegasus for abstractive summarization

<https://huggingface.co/SophieTr/fine-tune-Pegasus-large2>

text=In+einem+Krankenhaus+in+Canberra+haben+%C3%84rzte+einen+lebendigen%2C+acht+Zentimeter+langen+Rundwurm+aus+dem+Gehirn+einer+64-j%C3%A4hrigen+Frau+entfernt.+Bei+dem+Parasiten+handele+es+sich+um+die+Spezies+Ophidascaris+robertsi%2C+die+normalerweise+nur+in+Pythons+vorkomme%2C+hie%C3%9F+es+in+einer+neuen+Studie+im+Fachmagazin+Emerging+Infectious+Diseases.+Die+operierende+Neurochirurgin+sei+v%C3%B6llig+perplex+gewesen+und+habe+Kollegen+zurate+gezogen%2C+berichtete+der+australische+Guardian.+%22Dies+ist+der+erste+menschliche+Fall+von+Ophidascaris%2C+der+weltweit+beschrieben+wird%22%2C+zitierten+australische+Medien+Professor+Sanjaya+Senanayake%2C+einen+Experten+f%C3%BCr+Infektionskrankheiten+des+Canberra+Hospital+und+Mitautor+der+Studie.+%22Unseres+Wissens+ist+dies+auch+der+erste+Fall%2C+bei+dem+das+Gehirn+einer+S%C3%A4ugetierart%2C+ob+Mensch+oder+nicht%2C+betroffen+ist.%22+Die+%C3%84rzte+beschrieben+den+Wurm+als+%22lebendig+und+zappelnd%22.+Die+Patientin+aus+dem+Bundesstaat+New+South+Wales+hatte+Anfang+2021+zun%C3%A4chst+%C3%BCber+wochenlange+Bauchschmerzen+und+Durchfall+gefolgt+von+trockenem+Husten+und+Nachtschwei%C3%9F+geklagt.+Ein+Jahr+sp%C3%A4ter+geh%C3%B6rten+auch+Vergesslichkeit+und+Depressionen+zu+ihren+Symptomen.+Die+Frau+wurde+daraufhin+ins+Krankenhaus+von+Canberra+%C3%BCberwiesen.+Eine+MRT-Untersuchung+ihrer+Gehirns+ergab+schlie%C3%9Flich+Anomalien%2C+die+eine+Operation+n%C3%B6tig+machten.+%22Aber+die+Neurochirurgin+hat+bestimmt+niemals+gedacht%2C+dass+sie+einen+zappelnden+Wurm+finden+w%C3%BCrde%22%2C+sagte+Senanayake+dem+Gua

[rdian.+%22Neurochirurgen+haben+regelm%C3%A4%C3%9Fig+mit+Infektionen+im+Gehirn+zu+tun%2C+aber+das+war+ein+einmaliger+Fund+-+damit+hatte+niemand+gerechnet.%22.](#)

[```
model_name = "SophieTr/fine-tune-Pegasus-large"
tokenizer = PegasusTokenizer.from_pretrained\(model_name\)
model = PegasusForConditionalGeneration.from_pretrained\(model_name\)
```](https://huggingface.co/QuickRead/fine-tune-Pegasus?text=In+einem+Krankenhaus+in+Canberra+haben+%C3%84rzte+einen+lebendigen%2C+acht+Zentimeter+langen+Rundwurm+aus+dem+Gehirn+einer+64-j%C3%A4hrigen+Frau+entfernt.+Bei+dem+Parasiten+handele+es+sich+um+die+Spezies+Ophidascaris+robertsi%2C+die+normalerweise+nur+in+Pythons+vorkomme%2C+hie%C3%9F+es+in+einer+neuen+Studie+im+Fachmagazin+Emerging+Infectious+Diseases.+Die+operierende+Neurochirurgin+sei+v%C3%B6llig+perplex+gewesen+und+hab+Kollegen+zur+Zurückgezogen%2C+berichtet+der+australische+Guardian.+%22Dies+ist+der+erste+menschliche+Fall+von+Ophidascaris%2C+der+weltweit+beschrieben+wird%22%2C+zitierten+australische+Medien+Professor+Sajanya+Senanayake%2C+einen+Experten+f%C3%BCr+Infektionskrankheiten+des+Canberra+Hospital+und+Mitautor+der+Studie.+%22Unseres+Wissens+ist+dies+auch+der+erste+Fall%2C+bei+dem+das+Gehirn+einer+S%C3%A4ugetierart%2C+ob+Mensch+oder+nicht%2C+betroffen+ist.%22+Die+%C3%84rzte+beschrieben+den+Wurm+als+%22lebendig+und+zappelnd%22.+Die+Patientin+aus+dem+Bundesstaat+New+South+Wales+hatte+Anfang+2021+zun%C3%A4chst+%C3%BCber+wochenlange+Bauchschmerzen+und+Durchfall+gefolgt+von+trockenem+Husten+und+Nachtschweiß%C3%9F+geklagt.+Ein+Jahr+sp%C3%A4ter+geh%C3%B6rten+auch+Vergesslichkeit+und+Depressionen+zu+ihren+Symptomen.+Die+Frau+wurde+daraufhin+ins+Krankenhaus+von+Canberra+%C3%BCberwiesen.+Eine+MRT-Untersuchung+ihres+Gehirns+ergab+schlie%C3%9Flich+Anomalien%2C+die+eine+Operation+n%C3%B6tig+machten.+%22Aber+die+Neurochirurgin+hat+bestimmt+niemals+gedacht%2C+dass+sie+einen+zappelnden+Wurm+finden+w%C3%BCrde%22%2C+sagte+Senanayake+dem+Guardian.+%22Neurochirurgen+haben+regelm%C3%A4%C3%9Fig+mit+Infektionen+im+Gehirn+zu+tun%2C+aber+das+war+ein+einmaliger+Fund+-+damit+hatte+niemand+gerechnet.%22.</a></p>
</div>
<div data-bbox=)

|                                             |                                     |
|---------------------------------------------|-------------------------------------|
| Downloading spiece.model: 100%              | 1.91M/1.91M [00:00<00:00, 11.8MB/s] |
| Downloading (...)cial_tokens_map.json: 100% | 1.34k/1.34k [00:00<00:00, 53.5kB/s] |
| Downloading (...)okenizer_config.json: 100% | 1.60k/1.60k [00:00<00:00, 42.8kB/s] |
| Downloading (...)ve/main/config.json: 100%  | 3.12k/3.12k [00:00<00:00, 136kB/s]  |
| Downloading torch_model.bin: 100%           | 2.28G/2.28G [00:46<00:00, 57.5MB/s] |

```
def summarize(text):
 input_ids = tokenizer(text, return_tensors="pt").input_ids

 # Generate the output (Here, beam search was used however other strategy as wished may be used)
 output = model.generate(
 input_ids,
 max_length=150,
 min_length=50,
 num_beams=5,
 early_stopping=True)

 return tokenizer.decode(output[0], skip_special_tokens=True)

df['Abstractive Summarized Text (Germany)'] = df['TEXT'].apply(summarize)
df

df['Abstractive Summarized Text (English)'] = df['English Translated Text'].apply(summarize)
df
```

| ID  | TEXT                                              | source                                            | English Translated Text                           | Abstractive Summarized Text (Germany)             | Summarized Text (English)                         |
|-----|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| 0 1 | In einem Krankenhaus in Canberra haben Ärzte e... | https://www.sueddeutsche.de/wissen/australien-... | Doctors removed a live roundworm, three inches... | "Dies ist der erste menschliche Fall von Ophid... | "To our knowledge, this is also the first case... |
| 1 2 | Künstliche Intelligenz (KI) findet auch in der... | https://www.bild.de/ratgeber/gesundheit/gesund... | Artificial intelligence (AI) is also finding m... | Und: Die Hinttt der Dias ist manchmal nicht se... | Researchers at Harvard University have now dev... |
| 2 3 | Das landläufige Bild, das der Schriftsteller W... | https://www.jungewelt.de/artikel/457367.litera... | The common image that the writer Wolfgang Herr... | In seinen wenigen Interviews, die zeitlich mit... | Anyone who has read "This Side of the Van Alle... |
| 3 4 | In Kanada organisieren sich Drogenkonsument*in... | https://www.nd-aktuell.de/artikel/1175963.drog... | In Canada, drug users organize themselves in a... | Das ist auch deshalb mglich geworden, weil uns... | We started 25 years ago, at a time when many p... |
| 4 5 | taz: Frau Scharrel, immer wieder werden Frauen... | https://taz.de/Verbandschefin-ueber-Abtreibung... | taz: Ms. Scharrel, gynecologists who offer abo... | Seither kommen neue Gruppen oder Einzelpersone... | Scharrel, gynecologists who offer abortions in... |
| 5 6 | Wer Hundebesitzer im Umgang mit ihren Tieren      | https://www.tagesspiegel.de/gesundheit/von-       | Anyone who observes dog                           | Zwar sind auch potenziell sehr                    | Potentially very dangerous germs                  |
|     |                                                   |                                                   | Vaccines based on                                 | Er den wichtigsten                                | Promising data are                                |

## ▼ Extractive summarization without transformers

```

!pip install -U spacy
!python -m spacy download de_core_news_sm

Requirement already satisfied: spacy in /usr/local/lib/python3.10/dist-packages (3.6.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.0.11)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.0.0)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.0.9)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.0.7)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.0.8)
Requirement already satisfied: thinc<8.2.0,>=8.1.8 in /usr/local/lib/python3.10/dist-packages (from spacy) (8.1.12)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.1.2)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.4.7)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.0.9)
Requirement already satisfied: typer<0.10.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (0.9.0)
Requirement already satisfied: pathy<0.10.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (0.10.2)
Requirement already satisfied: smart-open<7.0.0,>=5.2.1 in /usr/local/lib/python3.10/dist-packages (from spacy) (6.3.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (4.66.1)
Requirement already satisfied: numpy>=1.15.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.23.5)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.31.0)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.6.4)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.1.2)
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages (from spacy) (67.7.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (23.1)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.3.0)
Requirement already satisfied: annotated-types>=0.4.0 in /usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (0.5.0)
Requirement already satisfied: pydantic-core==2.6.1 in /usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (2.6.1)
Requirement already satisfied: typing-extensions>=4.6.1 in /usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (4.6.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0) (3.2.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0) (2023.7.22)
Requirement already satisfied: blis<0.8.0,>=0.7.8 in /usr/local/lib/python3.10/dist-packages (from thinc<8.2.0,>=8.1.8) (0.7.10)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.10/dist-packages (from thinc<8.2.0,>=8.1.8) (0.0.4)
Requirement already satisfied: click<9.0.0,>=7.1.1 in /usr/local/lib/python3.10/dist-packages (from typer<0.10.0,>=0.3.0) (8.1.3)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (from Jinja2) (2.1.3)
2023-09-06 19:35:27.060059: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT
Collecting de-core-news-sm==3.6.0
 Downloading https://github.com/explosion/spacy-models/releases/download/de_core_news_sm-3.6.0/de_core_news_sm-3.6.0-py3-none-any.whl (14.6/14.6 MB) 23.0 MB/s eta 0:00:00
Requirement already satisfied: spacy<3.7.0,>=3.6.0 in /usr/local/lib/python3.10/dist-packages (from de-core-news-sm==3.6.0) (3.6.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (3.0.11)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (1.0.0)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (1.0.9)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (2.0.7)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (3.0.8)
Requirement already satisfied: thinc<8.2.0,>=8.1.8 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (8.1.12)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (1.1.2)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (2.4.7)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (2.0.9)
Requirement already satisfied: typer<0.10.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (0.9.0)
Requirement already satisfied: pathy<0.10.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (0.10.2)
Requirement already satisfied: smart-open<7.0.0,>=5.2.1 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (6.3.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (4.66.1)
Requirement already satisfied: numpy>=1.15.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (1.23.5)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (2.31.0)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0) (2.6.4)

```

```
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0->de-core-news-
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0->de-core-
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6.0->de-c
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.10/dist-packages (from spacy<3.7.0,>=3.6
Requirement already satisfied: annotated-types>=0.4.0 in /usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!>1.8)
```

```
import spacy
from string import punctuation
from heapq import nlargest
nlp = spacy.load('de_core_news_sm')
```

```
tokens = [token.text for token in doc] #Creating tokens from sentences
#print(tokens)
```

```
punctuation = punctuation + '\n'
punctuation
```

```
'!"#$%&\'()*+,-./:;<=>?@[\\]^_`{|}~\n'
```

### Cleaning token and enlisting in dictionary

```
word_frequencies = {}
for word in doc:
 if word.text.lower() not in punctuation:
 if word.text not in word_frequencies.keys():
 word_frequencies[word.text] = 1
 else:
 word_frequencies[word.text] += 1
```

```
word_frequencies
```

```
{'In': 1,
 'einem': 1,
 'Krankenhaus': 2,
 'in': 3,
 'Canberra': 3,
 'haben': 2,
 'Ärzte': 2,
 'einen': 3,
 'lebendigen': 1,
 'acht': 1,
 'Zentimeter': 1,
 'langen': 1,
 'Rundwurm': 1,
 'aus': 2,
 'dem': 5,
 'Gehirn': 3,
 'einer': 3,
 '64-jährigen': 1,
 'Frau': 2,
 'entfernt': 1,
 'Bei': 1,
 'Parasiten': 1,
 'handele': 1,
 'es': 2,
 'sich': 1,
 'um': 1,
 'die': 4,
 'Spezies': 1,
 'Ophidascaris': 2,
 'robertsi': 1,
 'normalerweise': 1,
 'nur': 1,
 'Pythons': 1,
 'vorkomme': 1,
 'hieß': 1,
 'neuen': 1,
 'Studie': 2,
 'im': 2,
 'Fachmagazin': 1,
 'Emerging': 1,
 'Infectious': 1,
 'Diseases': 1,
 'Die': 4,
 'operierende': 1,
 'Neurochirurgin': 2,
 'sei': 1,
 'völlig': 1,
 'perplex': 1,
 'gewesen': 1,
 'und': 6,
 'habe': 1,
 'Kollegen': 1,
 'zurate': 1,
```

```
'gezogen': 1,
'berichtete': 1,
'der': 5,
'australische': 2,
'Guardian': 2,
```

Seeing the words with the max frequency

```
max_frequency = max(word_frequencies.values())
max_frequency
```

6

Normalizing dictionary value

```
for word in word_frequencies.keys():
 word_frequencies[word] = word_frequencies[word]/max_frequency
```

```
word_frequencies
{'zunächst': 0.16666666666666666,
'über': 0.16666666666666666,
'wochenlange': 0.16666666666666666,
'Bauchschmerzen': 0.16666666666666666,
'Durchfall': 0.16666666666666666,
'gefolgt': 0.16666666666666666,
'trockenem': 0.16666666666666666,
'Husten': 0.16666666666666666,
'Nachtschweiß': 0.16666666666666666,
'geklagt': 0.16666666666666666,
'Ein': 0.16666666666666666,
'Jahr': 0.16666666666666666,
'später': 0.16666666666666666,
'gehörten': 0.16666666666666666,
'Vergesslichkeit': 0.16666666666666666,
'Depressionen': 0.16666666666666666,
'zu': 0.3333333333333333,
'ihren': 0.16666666666666666,
'Symptomen': 0.16666666666666666,
'wurde': 0.16666666666666666,
'daraufhin': 0.16666666666666666,
'ins': 0.16666666666666666,
'überwiesen': 0.16666666666666666,
'Eine': 0.16666666666666666,
'MRT-Untersuchung': 0.16666666666666666,
'ihres': 0.16666666666666666,
'Gehirns': 0.16666666666666666,
'ergab': 0.16666666666666666,
'schließlich': 0.16666666666666666,
'Anomalien': 0.16666666666666666,
'eine': 0.16666666666666666,
'Operation': 0.16666666666666666,
'nötig': 0.16666666666666666,
'machten': 0.16666666666666666,
'Aber': 0.16666666666666666,
'hat': 0.16666666666666666,
'bestimmt': 0.16666666666666666,
'niemals': 0.16666666666666666,
'gedacht': 0.16666666666666666,
'dass': 0.16666666666666666,
'sie': 0.16666666666666666,
'zappelnden': 0.16666666666666666,
'finden': 0.16666666666666666,
'würde': 0.16666666666666666,
'sagte': 0.16666666666666666,
'Neurochirurgen': 0.16666666666666666,
'regelmäßig': 0.16666666666666666,
'mit': 0.16666666666666666,
'Infektionen': 0.16666666666666666,
'tun': 0.16666666666666666,
'aber': 0.16666666666666666,
'war': 0.16666666666666666,
'ein': 0.16666666666666666,
'einmaliger': 0.16666666666666666,
'Fund': 0.16666666666666666,
'damit': 0.16666666666666666,
'niemand': 0.16666666666666666,
'gerechnet': 0.16666666666666666}
```

Sentnce tokenization

```
sentence_tokens = [sent for sent in doc.sents]
print(sentence_tokens)
```

[In einem Krankenhaus in Canberra haben Ärzte einen lebendigen, acht Zentimeter langen Rundwurm aus dem Gehirn einer 64-

## Sentence scores for detecting most important sentence

```
sentence_scores = {}
for sent in sentence_tokens:
 for word in sent:
 if word.text.lower() in word_frequencies.keys():
 if sent not in word_frequencies.keys():
 sentence_scores[sent]=word_frequencies[word.text.lower()]
 else:
 sentence_scores[sent]+=word_frequencies[word.text.lower()]
sentence_scores

{In einem Krankenhaus in Canberra haben Ärzte einen lebendigen, acht Zentimeter langen Rundwurm aus dem Gehirn einer 64-jährigen Frau entfernt.: 0.16666666666666666,
Bei dem Parasiten handele es sich um die Spezies Ophidascaris robertsi, die normalerweise nur in Pythons vorkomme, hieß es in einer neuen Studie im Fachmagazin Emerging Infectious Diseases.: 0.3333333333333333,
Die operierende Neurochirurgin sei völlig perplex gewesen und habe Kollegen zurate gezogen, berichtete der australische Guardian.: 0.3333333333333333,
"Dies ist der erste menschliche Fall von Ophidascaris, der weltweit beschrieben wird", zitierten australische Medien Professor Sanjaya Senanayake, einen Experten für Infektionskrankheiten des Canberra Hospital und Mitautor der Studie.: 0.8333333333333334,
"Unseres Wissens ist dies auch der erste Fall, bei dem das Gehirn einer Säugetierart, ob Mensch oder nicht, betroffen ist.": 0.5,
Die Ärzte beschrieben den Wurm als "lebendig und zappelnd".: 0.16666666666666666,
Die Patientin aus dem Bundesstaat New South Wales hatte Anfang 2021 zunächst über wochenlange Bauchschmerzen und Durchfall gefolgt von trockenem Husten und Nachtschweiß geklagt.: 0.16666666666666666,
Ein Jahr später gehörten auch Vergesslichkeit und Depressionen zu ihren Symptomen.: 0.16666666666666666,
Die Frau wurde daraufhin ins Krankenhaus von Canberra überwiesen.: 0.16666666666666666,
Eine MRT-Untersuchung ihres Gehirns ergab schließlich Anomalien, die eine Operation nötig machten.: 0.16666666666666666,
"Aber die Neurochirurgin hat bestimmt niemals gedacht, dass sie einen zappelnden Wurm finden würde", sagte Senanayake dem Guardian.: 0.8333333333333334,
"Neurochirurgen haben regelmäßig mit Infektionen im Gehirn zu tun, aber das war ein einmaliger Fund - damit hatte niemand gerechnet.: 0.16666666666666666}
```

```
select_length = int(len(sentence_tokens)*0.3) #The summary will have only the 30% of the main text
select_length
```

3

```
summary = nlargest(select_length,sentence_scores,key = sentence_scores.get)
#summary
```

```
["Dies ist der erste menschliche Fall von Ophidascaris, der weltweit beschrieben wird", zitierten australische Medien Professor Sanjaya Senanayake, einen Experten für Infektionskrankheiten des Canberra Hospital und Mitautor der Studie.,
"Aber die Neurochirurgin hat bestimmt niemals gedacht, dass sie einen zappelnden Wurm finden würde", sagte Senanayake dem Guardian.,
"Unseres Wissens ist dies auch der erste Fall, bei dem das Gehirn einer Säugetierart, ob Mensch oder nicht, betroffen ist."]
```

```
sent = ""
for i in summary:
 sent = f"{sent} {i}"
```

```
sent
```

```
new_str = sent.replace("'", " ")
new_str = new_str[2:-1]
new_str
```

```
' Dies ist der erste menschliche Fall von Ophidascaris, der weltweit beschrieben wird", zitierten australische Medien Professor Sanjaya Senanayake, einen Experten für Infektionskrankheiten des Canberra Hospital und Mitautor der Studie. "Aber die Neurochirurgin hat bestimmt niemals gedacht, dass sie einen zappelnden Wurm finden würde", sagte Senanayake dem Guardian. "Unseres Wissens ist dies auch der erste Fall, bei dem das Gehirn einer Säugetierart, ob Mensch oder nicht, betroffen ist. "
```

```
def extractive_summarize(text):
```

```
 from string import punctuation
```

```
 doc = nlp(text)
 tokens = [token.text for token in doc] #Creating tokens from sentences
 punctuation = punctuation + '\n'
```

```

word_frequencies = {} #Making a dictionary and their frequencies
for word in doc:
 if word.text.lower() not in punctuation:
 if word.text not in word_frequencies.keys():
 word_frequencies[word.text] = 1
 else:
 word_frequencies[word.text]+=1

max_frequency = max(word_frequencies.values()) #Seeing the word with the max frequency in the dictionary

for word in word_frequencies.keys():
 word_frequencies[word] = word_frequencies[word]/max_frequency #Normalizing dictionary value

sentence_tokens = [sent for sent in doc.sents] #Creating sentence tokens

sentence_scores = {} #Creating sentence scores to determine the most important sentences
for sent in sentence_tokens:
 for word in sent:
 if word.text.lower() in word_frequencies.keys():
 if sent not in word_frequencies.keys():
 sentence_scores[sent]=word_frequencies[word.text.lower()]
 else:
 sentence_scores[sent]+=word_frequencies[word.text.lower()]

select_length = int(len(sentence_tokens)*0.3) #The summary will have only the 30% of the main text

summary = nlargest(select_length,sentence_scores,key = sentence_scores.get) #list of the sentences of the main summary

sent = ""
for i in summary:
 sent = f"{sent} {i}"

new_str = sent.replace("'", " ")
new_str = new_str[2:-1]
#print(new_str)
return new_str

df = df[["ID","TEXT", "source", "English Translated Text", "Abstractive Summarized Text (Germany)", "Summarized Text (English)"]]
df

```

| ID  | TEXT                                              | source                                                                                                            | English Translated Text                            | Abstractive Summarized Text (Germany)              | Summarized Text (English)                         |
|-----|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---------------------------------------------------|
| 0 1 | In einem Krankenhaus in Canberra haben Ärzte e... | <a href="https://www.sueddeutsche.de/wissen/australien-...">https://www.sueddeutsche.de/wissen/australien-...</a> | Doctors removed a live roundworm, three inches...  | "Dies ist der erste menschliche Fall von Ophid...  | "To our knowledge, this is also the first case... |
| 1 2 | Künstliche Intelligenz (KI) findet auch in der... | <a href="https://www.bild.de/ratgeber/gesundheit/gesund...">https://www.bild.de/ratgeber/gesundheit/gesund...</a> | Artificial intelligence (AI) is also finding m...  | Und: Die Hinttt der Dias ist manchmal nicht se...  | Researchers at Harvard University have now dev... |
| 2 3 | Das landläufige Bild, das der Schriftsteller W... | <a href="https://www.jungewelt.de/artikel/457367.litera...">https://www.jungewelt.de/artikel/457367.litera...</a> | The common image that the writer Wolfgang Herr...  | In seinen wenigen Interviews, die zeitlich mit...  | Anyone who has read "This Side of the Van Alle... |
| 3 4 | In Kanada organisieren sich Drogenkonsument*in... | <a href="https://www.nd-aktuell.de/artikel/1175963.drog...">https://www.nd-aktuell.de/artikel/1175963.drog...</a> | In Canada, drug users organize themselves in a...  | Das ist auch deshalb mglich geworden, weil uns...  | We started 25 years ago, at a time when many p... |
| 4 5 | taz: Frau Scharrel, immer wieder werden Frauen... | <a href="https://taz.de/Verbandschefin-ueber-Abtreibung...">https://taz.de/Verbandschefin-ueber-Abtreibung...</a> | taz: Ms. Scharrel, gynecologists who offer abo...  | Seither kommen neue Gruppen oder Einzelpersone...  | Scharrel, gynecologists who offer abortions in... |
| 5 6 | Wer Hundebesitzer im Umgang mit ihren Tieren b... | <a href="https://www.tagesspiegel.de/gesundheit/von-weg...">https://www.tagesspiegel.de/gesundheit/von-weg...</a> | Anyone who observes dog owners interacting wit...  | Zwar sind auch potenziell sehr gefährliche Keim... | Potentially very dangerous germs are also know... |
| 6 7 | Impfstoffe auf Basis von mRNA brachten im Kamp... | <a href="https://www.handelsblatt.com/technik/medizin/b...">https://www.handelsblatt.com/technik/medizin/b...</a> | Vaccines based on mRNA brought the breakthrough... | Fr den wichtigsten Hoffnungstrger, einen Hautk...  | Promising data are now available for the most ... |
| 7 8 | Für Frauen zwischen 50 und 69 Jahren gehört es... | <a href="https://www.welt.de/gesundheit/article24699962...">https://www.welt.de/gesundheit/article24699962...</a> | For women between the ages of 50 and 69 it is ...  | Fr Frauen zwischen 50 und 69 Jahren gehrt es z...  | In the future, women between the ages of 70 an... |
|     | Die Zahl der neu                                  |                                                                                                                   | The number of                                      | Dabei muss nach                                    | The number of                                     |

```
df['Extractive Summarized Text (Germany)'] = df['TEXT'].apply(extractive_summarize)
df
```

Dies ist der erste menschliche Fall von Ophidascaris, der weltweit beschrieben wird , zitierten australische Medien Profet  
etzt haben Forscher der Universität Harvard eine KI entwickelt, mit der die Behandlung von Gehirn-Tumoren in Zukunft schr  
ie Kritik (die diesen Autor wahrlich nicht schlecht behandelt hat) sei korrupt, zu Lesungen raffte er sich nur »wegen der  
ch selbst bin an einer Gruppe beteiligt, die sich Vancouver Area Network of Drug Users nennt, kurz Vandu. Konkret meint  
ie ist die Lage in Schleswig-Holstein? In Schleswig-Holstein erlebten wir Proteste vor den Praxen nur in Einzelfällen. Si  
nd das tun die Keime tatsächlich auch. Und während Hunde bei Erwachsenen vor allem in Hand und Wade beißen, ist es bei kl  
as zeigen detaillierte Daten aus einer Phase-2-Studie der US-Biotech-Firma Moderna und ihres Partners Merck & Co. zu dem  
ie Altersgrenze solle nach oben ausgeweitet werden, sagte eine Sprecherin des Gemeinsamen Bundesausschusses (G-BA). Vorat  
abei muss nach Angaben des RKI allerdings berücksichtigt werden, dass die Zahl der HIV-Neuinfektionen 2020 und 2021 deutl  
<ipython-input-73-aa63512038e5>:1: SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: [https://pandas.pydata.org/pandas-docs/stable/user\\_guide/indexing.html#returning-a-slice-of-a-df-or-df-view](https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-slice-of-a-df-or-df-view)  
df['Extractive Summarized Text (Germany)'] = df['TEXT'].apply(extractive\_summarize)

| ID | TEXT | source                                            | English Translated Text                                                                                           | Abstractive Summarized Text (Germany)             | Summarized Text (English)                          | Extractive Summarized Text (Germany)              |                                                   |
|----|------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| 0  | 1    | In einem Krankenhaus in Canberra haben Ärzte e... | <a href="https://www.sueddeutsche.de/wissen/australien-...">https://www.sueddeutsche.de/wissen/australien-...</a> | Doctors removed a live roundworm, three inches... | "Dies ist der erste menschliche Fall von Ophid...  | "To our knowledge, this is also the first case... | Dies ist der erste menschliche Fall von Ophida... |
| 1  | 2    | Künstliche Intelligenz (KI) findet auch in der... | <a href="https://www.bild.de/ratgeber/gesundheit/gesund...">https://www.bild.de/ratgeber/gesundheit/gesund...</a> | Artificial intelligence (AI) is also finding m... | Und: Die Hintt der Dias ist manchmal nicht se...   | Researchers at Harvard University have now dev... | etzt haben Forscher der Universität Harvard ei... |
| 2  | 3    | Das landläufige Bild, das der Schriftsteller W... | <a href="https://www.jungewelt.de/artikel/457367.litera...">https://www.jungewelt.de/artikel/457367.litera...</a> | The common image that the writer Wolfgang Herr... | In seinen wenigen Interviews, die zeitlich mit...  | Anyone who has read "This Side of the Van Alle... | ie Kritik (die diesen Autor wahrlich nicht sch... |
| 3  | 4    | In Kanada organisieren sich Drogenkonsument*in... | <a href="https://www.nd-aktuell.de/artikel/1175963.drog...">https://www.nd-aktuell.de/artikel/1175963.drog...</a> | In Canada, drug users organize themselves in a... | Das ist auch deshalb mglich geworden, weil uns...  | We started 25 years ago, at a time when many p... | ch selbst bin an einer Gruppe beteiligt, die s... |
| 4  | 5    | taz: Frau Scharrel, immer wieder werden Frauen... | <a href="https://taz.de/Verbandschefin-ueber-Abtreibung...">https://taz.de/Verbandschefin-ueber-Abtreibung...</a> | taz: Ms. Scharrel, gynecologists who offer abo... | Seither kommen neue Gruppen oder Einzelpersone...  | Scharrel, gynecologists who offer abortions in... | ie ist die Lage in Schleswig-Holstein? In Schl... |
| 5  | 6    | Wer Hundebesitzer im Umgang mit ihren Tieren b... | <a href="https://www.tagesspiegel.de/gesundheit/von-weg...">https://www.tagesspiegel.de/gesundheit/von-weg...</a> | Anyone who observes dog owners interacting wit... | Zwar sind auch potenziell sehr gefährliche Keim... | Potentially very dangerous germs are also know... | nd das tun die Keime tatsächlich auch. Und wäh... |
| 6  | 7    | Impfstoffe auf Basis von mRNA brachten im Kamp... | <a href="https://www.handelsblatt.com/technik/medizin/b...">https://www.handelsblatt.com/technik/medizin/b...</a> | Vaccines based on mRNA brought the                | Fr den wichtigsten Hoffnungstrger,                 | Promising data are now available for              | as zeigen detaillierte Daten aus einer Phase-     |

```
df['Extractive Summarized Text (English)'] = df['English Translated Text'].apply(extractive_summarize)
df
```

ut since paragraph 219a of the Criminal Code What is the situation in Schleswig-Holstein? rise up morally and berate gynecologists. The more crucial risk is kissing the As After all, dogs stick their So is it dangerous? bacterium is particularly dangerous. Study by the Moderna and Merck confirmed their plans to test the drug in other types of cancer. Promising data are now available. The age limit should be extended upwards, said a spokeswoman for the Federal Joint Committee (G-BA). On behalf of the legislators. Around 2,500 infections were reported last year, as the Robert Koch Institute (RKI) are not actual new diagnoses, but first-time infections.

| ID | TEXT | source                                            | English Translated Text                           | Abstractive Summarized Text (Germany)             | Summarized Text (English)                          | Extractive Summarized Text (Germany)              | Extractive Summarized Text (English)              |
|----|------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| 0  | 1    | In einem Krankenhaus in Canberra haben Ärzte e... | https://www.sueddeutsche.de/wissen/australien-... | Doctors removed a live roundworm, three inches... | "Dies ist der erste menschliche Fall von Ophid...  | "To our knowledge, this is also the first case... | Dies ist der erste menschliche Fall von Ophida... |
| 1  | 2    | Künstliche Intelligenz (KI) findet auch in der... | https://www.bild.de/ratgeber/gesundheit/gesund... | Artificial intelligence (AI) is also finding m... | Und: Die Hintt der Dias ist manchmal nicht se...   | Researchers at Harvard University have now dev... | etzt haben Forscher der Universität Harvard ei... |
| 2  | 3    | Das landläufige Bild, das der Schriftsteller W... | https://www.jungewelt.de/artikel/457367.litera... | The common image that the writer Wolfgang Herr... | In seinen wenigen Interviews, die zeitlich mit...  | Anyone who has read "This Side of the Van Alle... | ie Kritik (die diesen Autor wahrlich nicht sch... |
| 3  | 4    | In Kanada organisieren sich Drogenkonsument*in... | https://www.nd-aktuell.de/artikel/1175963.drog... | In Canada, drug users organize themselves in a... | Das ist auch deshalb möglich geworden, weil uns... | We started 25 years ago, at a time when many p... | ch selbst bin an einer Gruppe beteiligt, die s... |
|    |      |                                                   |                                                   | taz: Ms.                                          | Seither                                            | Scharrel                                          | ie ist die                                        |
|    |      | Frauen...                                         |                                                   | WHO offer                                         | Einzelperson                                       | Abstraktionen in                                  | Holstein? In                                      |

➤ Using DISTILBART for extractive summarization

<https://huggingface.co/sshleifer/distilbart-cnn-12-6>

```
wit... also know... wäh... ...

!pip install transformers

Requirement already satisfied: transformers in /usr/local/lib/python3.10/dist-packages (4.33.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from transformers) (3.12.2)
Requirement already satisfied: huggingface-hub<1.0,>=0.15.1 in /usr/local/lib/python3.10/dist-packages (from transformers) (0.15.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.10/dist-packages (from transformers) (1.23.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from transformers) (23.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.10/dist-packages (from transformers) (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.10/dist-packages (from transformers) (2023.6.3)
Requirement already satisfied: requests in /usr/local/lib/python3.10/dist-packages (from transformers) (2.31.0)
Requirement already satisfied: tokenizers!=0.11.3,<0.14,>=0.11.1 in /usr/local/lib/python3.10/dist-packages (from transformers) (0.13.3)
Requirement already satisfied: safetensors>=0.3.1 in /usr/local/lib/python3.10/dist-packages (from transformers) (0.3.3)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.10/dist-packages (from transformers) (4.66.1)
Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages (from huggingface-hub<1.0,>=0.15.1->transformers) (2023.6.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.10/dist-packages (from huggingface-hub<1.0,>=0.15.1->transformers) (4.5.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests->transformers) (3.2.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests->transformers) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests->transformers) (2.0.4)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests->transformers) (2023.7.22)

import transformers

to_tokenize = df['English Translated Text'][0]
to_tokenize

'Doctors removed a live roundworm, three inches long, from the brain of a 64-year-old woman at a Canberra hospital. The parasite is the species Ophidascaris robertsi, which is normally found only in pythons, according to a new study in the journal Emerging Infectious Diseases. The neurosurgeon who operated on was completely perplexed and consulted colleagues, the Australian Guardian reported. "This is the first human case of Ophidascaris to be described anywhere in the world," Australian media quoted Professor Sanjaya Senanayake, an infectious disease expert at Canberra Hospital and co-author of the study as saying. "To our knowledge, this is also the first case in which the brain of a mammalian species, whether human or not, is affected." Doctors described the worm as "alive and wriggling." In early 2021, the patient from the state of New South Wales initially complained of weeks of abdominal pain and diarrhea, followed by a dry cough and night sweats. A year later her symptoms also included...'

summarizer = pipeline("summarization")

summarized = summarizer(to_tokenize, min_length=75, max_length=300)
```

No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (<https://huggingface.co/sshleifer/>)  
Using a pipeline without specifying a model name and revision in production is not recommended.

Downloading (...)lve/main/config.json: 100% 1.80k/1.80k [00:00<00:00, 21.7kB/s]

Downloading pytorch\_model.bin: 100% 1.22G/1.22G [00:16<00:00, 102MB/s]

Downloading (...)okenizer\_config.json: 100% 26.0/26.0 [00:00<00:00, 781B/s]

Downloading (...)olve/main/vocab.json: 100% 899k/899k [00:00<00:00, 7.06MB/s]

Downloading (...)olve/main/merges.txt: 100% 456k/456k [00:00<00:00, 15.9MB/s]

Your max\_length is set to 300, but your input\_length is only 298. Since this is a summarization task, where outputs short  
[{'summary\_text': ' A 64-year-old Australian woman was diagnosed with abdominal pain and diarrhea . She complained of for

```
print(summarized[0]['summary_text'])
```

A 64-year-old Australian woman was diagnosed with abdominal pain and diarrhea . She complained of forgetfulness and dep

```
summ=' '.join([str(i) for i in summarized])
summl=len(summ.split())
```

```
print(" The no. of words in the summarization is :"+str(summl))
```

The no. of words in the summarization is :60

```
def extractive_sum_trans(text):
 to_tokenize = text
```

```
 summarizer = pipeline("summarization")
```

```
 summarized = summarizer(to_tokenize, min_length=75, max_length=300)
```

```
 sent = summarized[0]['summary_text']
```

```
 return sent
```

```
df['Extractive Summarized Text using transformers (Germany)'] = df['TEXT'].apply(extractive_sum_trans)
df
```

|   |   |                                                         |                                                                                                                         |                                                                  |        |
|---|---|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------|
| 0 | 1 | In einem Krankenhaus<br>in Canberra haben<br>Ärzte e... | <a href="https://www.sueddeutsche.de/wissen/australien-...">https://www.sueddeutsche.de/wissen/australien-...</a>       | Doctors<br>removed a<br>live<br>roundworm,<br>three<br>inches... |        |
| 1 | 2 | Künstliche Intelligenz<br>(KI) findet auch in<br>der... | <a href="https://www.bild.de/ratgeber/gesundheit/gesund...">https://www.bild.de/ratgeber/gesundheit/gesund...</a>       | Artificial<br>intelligence<br>(AI) is also<br>finding m...       | l<br>m |
| 2 | 3 | Das landläufige Bild,<br>das der Schriftsteller<br>W... | <a href="https://www.jungewelt.de/artikel/457367.litera...">https://www.jungewelt.de/artikel/457367.litera...</a>       | The common<br>image that<br>the writer<br>Wolfgang<br>Herr...    |        |
| 3 | 4 | In Kanada<br>organisieren sich<br>Drogenkonsument*in... | <a href="https://www.nd-aktuell.de/artikel/1175963.drog...">https://www.nd-aktuell.de/artikel/1175963.drog...</a>       | In Canada,<br>drug users<br>organize<br>themselves in<br>a...    | c<br>c |
| 4 | 5 | taz: Frau Scharrel,<br>immer wieder werden<br>Frauen... | <a href="https://taz.de/Verbandschefin-ueber-Abtreibung...">https://taz.de/Verbandschefin-ueber-Abtreibung...</a>       | taz: Ms.<br>Scharrel,<br>gynecologists<br>who offer<br>abo...    | Ei     |
| 5 | 6 | Wer Hundebesitzer im<br>Umgang mit ihren<br>Tieren b... | <a href="https://www.tagesspiegel.de/gesundheit/von-...weg...">https://www.tagesspiegel.de/gesundheit/von-...weg...</a> | Anyone who<br>observes dog<br>owners<br>interacting<br>wit       | Z<br>I |

```
df['Extractive Summarized Text using transformers (English)'] = df['English Translated Text'].apply(extractive_sum_trans)
df
```

|                                        |   |                                                         |                                                                                                                   |                                                                  |        |
|----------------------------------------|---|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------|
| 0                                      | 1 | In einem Krankenhaus<br>in Canberra haben<br>Ärzte e... | <a href="https://www.sueddeutsche.de/wissen/australien-...">https://www.sueddeutsche.de/wissen/australien-...</a> | Doctors<br>removed a<br>live<br>roundworm,<br>three<br>inches... |        |
| 1                                      | 2 | Künstliche Intelligenz<br>(KI) findet auch in<br>der... | <a href="https://www.bild.de/ratgeber/gesundheit/gesund...">https://www.bild.de/ratgeber/gesundheit/gesund...</a> | Artificial<br>intelligence<br>(AI) is also<br>finding m...       | l<br>m |
| 2                                      | 3 | Das landläufige Bild,<br>das der Schriftsteller<br>W... | <a href="https://www.jungewelt.de/artikel/457367.litera...">https://www.jungewelt.de/artikel/457367.litera...</a> | The common<br>image that<br>the writer<br>Wolfgang<br>Herr...    |        |
| 3                                      | 4 | In Kanada<br>organisieren sich<br>Drogenkonsument*in... | <a href="https://www.nd-aktuell.de/artikel/1175963.drog...">https://www.nd-aktuell.de/artikel/1175963.drog...</a> | In Canada,<br>drug users<br>organize<br>themselves in<br>a...    | c<br>c |
| 4                                      | 5 | taz: Frau Scharrel,<br>immer wieder werden<br>Frauen... | <a href="https://taz.de/Verbandschefin-ueber-Abtreibung...">https://taz.de/Verbandschefin-ueber-Abtreibung...</a> | taz: Ms.<br>Scharrel,<br>gynecologists<br>who offer<br>abo...    | Ei     |
| 5                                      | 6 | Wer Hundebesitzer im<br>Umgang mit ihren<br>Tieren b... | <a href="https://www.tagesspiegel.de/gesundheit/von-...">https://www.tagesspiegel.de/gesundheit/von-...</a>       | Anyone who<br>observes dog<br>owners<br>interacting<br>wit...    | Z<br>I |
| df.to_excel("summarized_results.xlsx") |   |                                                         |                                                                                                                   | Vaccines<br>based on                                             |        |
| 7                                      | 8 | Für Frauen zwischen<br>50 und 69 Jahren<br>gehört es... | <a href="https://www.welt.de/gesundheit/article24699962...">https://www.welt.de/gesundheit/article24699962...</a> | For women<br>between the<br>ages of 50<br>and 69 it is ...       | I      |
| 8                                      | 9 | Die Zahl der neu<br>gemeldeten HIV-Fälle<br>in Deuts... | <a href="https://www.faz.net/aktuell/wissen/medizin-ern...">https://www.faz.net/aktuell/wissen/medizin-ern...</a> | The number<br>of newly<br>reported HIV<br>cases in<br>Germ...    | I<br>a |